

# Review on AI and Knowledge Graphs for Crime Investigation

**Abstract**—“AI is getting much better at understanding human data, and this is leading to various new ways to investigate digital crime.”. Pairing powerful AI (like LLMs) with smart, context-aware reasoning could be a game-changer for how police analyze, understand, and use crime data. In this review, we demonstrate a survey of the existing approaches available in the research literature and AI-enhanced capabilities for complex crime-related data analysis, knowledge graph development, and intelligent question-answering systems to assist investigators. This review examines earlier works on entity extraction, graph-based reasoning, and retrieval-augmented architecture; and reports on their contributions and limitations in working with communities of practice, forensic datasets, and legal data. We’ll show that by teaming up a smart AI with a reliable database, the system gets better at understanding the full picture, makes fewer mistakes, and provides clear answers that investigators can actually explain and trust. This review emphasizes important gaps such as interoperability of a heterogeneous crime data representation, standardization of crime-related knowledge, and scalability of methods to apply in domains such as crime. This paper outlines future research opportunities in multi-source knowledge interoperability, building trustworthy and explainable AI systems, and developing an operational definition of real-time query optimization across AI models in modern crime investigation systems to enhance accuracy, reliability, and transparency.

## I. INTRODUCTION

The increasing complexity and volume of digital evidence present in the current investigations has led to requests for an intelligently automated systems that would assist agency personnel in analyzing, interpreting and associating of crime data. Now, investigators don’t need to painstakingly review the evidence, sorting through various independent, unlinked databases, and reviewing only what they know from past experience. Not only this method is slow, it can produce cost errors. The recent literature has covered various methods for examining crime data to solve this problem, particularly through using automated Artificial Intelligence (AI) methods, such as Large Language Models (LLMs) and Knowledge Graphs (KGs) that help in the understanding of crime data and facilitate context-aware reasoning.

LLMs like GPT and BERT have given unexpected competition in natural language processing, information retrieval, and context generation. However, when LLMs are deployed without any proper customized responsiveness in investigative principles, people should remain cautious about the limitations of directly applying LLMs due to unauthorized proofs for facts, hallucination behavior, and interpretable meanings. To solve limitations, Knowledge Graph-based reasoning frameworks have implemented as a complementary, alternative way in which entities (e.g., persons, places, organizations, events) and their relationships can be arranged into a knowledge base.

A future research direction is to incorporate LLMs with graph-based retrieval (RAG) mechanisms as a good way to build a data-driven and explainable assistive crime investigation system.

This article gives a thorough overview of the literature on AI-powered criminal analysis, highlighting how systems have developed based on natural language understanding, entity extraction, graph-queries, and reasoning features. The paper solves previous systems addressed challenges with mixed data, multiple data sources combination and context inference, and where failed with scalability, trust and domain-specific accommodation.

In addition, this study gives the problems of published systems and gives future research that is necessary to develop future LLM-KG hybrid architectures. A future LLM-KG hybrid architecture is used to improve the precision and efficiency of investigative work while can be explainable, usable, and can be operable—key factors for deployment in a real-world forensic context. Consider this review a starting point for anyone hoping to create the next generation of AI investigation tools—ones that are not only intelligent but also understandable and reliable.

## II. LITERATURE SURVEY

This section provides an extensive assessment of current research on the fusion of recent developments in integrating large language models (LLMs), knowledge graphs (KGs), and reasoning systems for intelligent question answering and data analysis. This section reviews numerous journal and conference papers focused on KG construction, ontology generation, retrieval-augmented generation (RAG), and hybrid reasoning systems. Each paper is evaluated to understand its methodology, contribution, and relevance in the development of explainable AI technologies for contemporary crime investigation system.

### A. Knowledge Graph Question Answering with Pre-Trained Language Models: Base paper

This paper examined the performance of pre-trained language models in answering factual questions over knowledge graphs. The researchers tested transformer-based models, such as BERT and RoBERTa, on datasets with both single-hop and multi-hop factual questions. They found these models are able to accurately answer simple questions about single entities, but there is a decline in performance quickly when reasoning or linking between entities is required to answer the question. The analysis pointed out that LLMs sometimes interpret the relationship structure of knowledge graphs incorrectly when explicit graph context is not provided. More importantly,

the paper highlights that using structured KG retrieval with language models will largely impact facts accuracy, reasoning, and interpretability, all of which are important characteristics for crime-investigation assistants.

### B. Retrieval-Augmented Generation survey

In this review, the authors provide an overview of hybrid retrieval-augmented generation (RAG) architectures that involve retrieval-augmented generative large language models (LLMs), retrievers, and generative LLM. These ways are designed to address imaginations and build deeper levels of facts understanding. The authors differentiate among various types of retrievers (dense vector, sparse, and structured DB/graph) and index/embedding ways, retrieval-to-prompt ways and evaluation ways for grounded generation. As they note, there are important tradeoffs to consider: vector retrieval typically involves semantic match capability and scalability while structured (DB/graph), precision and relational context; So, this can yield considerable design decisions in terms of what might be important in investigative domains, particularly where relational provenance is important. The survey covers hybrid retrievers as well as recent work on retrieval calibration and trust scoring, and also notes open challenges: multimodal retrieval, dynamic updates, and assessing solid evaluation metrics for real-world, high-stakes domains.

### C. Experimental analysis of LLMs in crime tasks

In the present experimental research, the authors provide a benchmark on LLMs (BART, GPT-3, GPT-4 series) on crime-focused, classification and prediction tasks, through implementations of zero-shot and few-shot prompting including iterative refinement. The authors suggest that large generative models show potential to perform better than your typical supervised models on conversational and classification-type tasks when the prompts are correctly constructed. However, LLMs still show a lack of consistency in factual accuracy and don't support structured multi-hop queries natively. The authors make the case LLMs are immensely powerful when completing tasks related to summarization, and exploratory analysis, but if you need to reach a conclusion after an investigation, then these models need to be supplemented with structured retrieval (RAG) or over with symbols. They also report limited replicability across datasets and call for standardization of evaluation datasets and metrics for policing contexts also.

### D. Evaluation of Large Language Models in Ontology and Knowledge Graph Creation

A new study assessed the performance of large language models like GPT-3.5, GPT-4, and Perplexity AI to build ontologies and knowledge graphs from unstructured text. Their experimental design included the evaluation of accuracy when these models were instructed to extract entities, define relations, and build domain hierarchies using carefully formed prompts. The author found that LLMs could automatically create coherent graph-based structures to an acceptable level

of accuracy but frequently would produce outputs that did not maintain consistency across domain and range, or hierarchies of subclasses. They concluded that while LLMs automates some manual labor when creating KGs, having a human supervisor and tuned prompts for a particular domain is necessary for accuracy in specialized domains like representing knowledge in criminal law.

### E. Hybrid SPARQL and Keyword Search for Knowledge Graph Question Answering

Another contribution to the journal put forth the idea of merging SPARQL query translation with keyword search for improved natural-language question answering over knowledge graphs. The authors accessed user queries in two separate paths—semantic keyword extraction to span the potential entities, and structured SPARQL generation to identify relations in a retrieved entity list. This new combined approach results in better recall and precision as compared to baselines with only one authoring method. Furthermore, the authors say this hybrid search method is the perfect 'translator' between structured databases and plain English. This is incredibly useful for AI assistants, as they must be able to understand an investigator's free-form question and immediately find the correct link in a complex web of evidence.

### F. Comparison of Literature

Table I summarizes major contributions, highlighting their sources, methods, limitations, and findings.

## III. LIMITATIONS OF EXISTING WORK

Although research to date has made meaningful strides in combining LLMs with knowledge graphs to enhance structured reasoning, there are still significant practical and theoretical limitations. In the following section, we critically discuss the key constraints identified in previous studies, including: limitations in scalability, limited explainability, overdependence on sources, and limited applicability to investigative scenarios. Understanding these limitations gives context in supporting the foundation for identifying specific research directions to address previously noted limitations, as well as enrich the AI and investigative opportunity.

### A. Limitations in Simple Knowledge Graph Question Answering

While effective for basic knowledge-graph question answering, this work is limited to single-hop reasoning and benchmark-style datasets. It does not consider noise in or incompleteness of the graphs, nor does it have any systems for dynamic updating or cross-domain adaptation. It also relies on the data used for pre-training and does not explain incorrect answers. As such, it is not suited for real-world investigative scenarios that require transparency and multi-step reasoning.

TABLE I  
COMPARISON OF RELATED WORKS

| S.No | Title                                                                  | Source / Year                                                  | Methods / Techniques Used                                                                | Key Findings                                                                                                                    |
|------|------------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 1    | Knowledge Graph Question Answering with Pre-Trained Language Models    | Hu et al., World Wide Web Journal, 2023                        | Fine-tuning of BERT and RoBERTa models for KGQA using single-hop and multi-hop datasets. | Performs well on simple queries but fails on multi-hop reasoning and lacks interpretability; requires structured graph support. |
| 2    | Evaluation of LLMs in Ontology and KG Creation                         | Dutta et al., Journal of Information and Knowledge, 2025       | GPT-3.5, GPT-4, Perplexity AI; prompt-based ontology and relation extraction.            | Automates KG generation but often misses hierarchical consistency; requires domain-specific prompting and human verification.   |
| 3    | Knowledge Graph Combined with Retrieval-Augmented Generation           | Wang and Shi, Academic Journal of Science and Technology, 2024 | Hybrid retrieval combining vector search and graph-based reasoning.                      | Improves factual accuracy and reduces hallucination; limited by lack of standardized evaluation and explainability.             |
| 4    | Knowledge Graph-Based QA Systems and Their Domain Applications         | Cao, Ni, and Dai, Academic Journal of Computing, 2021          | Entity linking, SPARQL querying, and semantic parsing.                                   | Works well for static structured data; struggles with scalability, ambiguity, and multi-domain data integration.                |
| 5    | Hybrid SPARQL and Keyword Search for Knowledge Graph QA                | Hu, Duan, and Dang, Knowledge and Information Systems, 2021    | Combines keyword search and SPARQL query templates for QA.                               | Enhances recall and precision but lacks deep semantic understanding and contextual ranking.                                     |
| 6    | TQAgent: Table-Based Question Answering Enhanced with Knowledge Graphs | Zhao et al., Applied Sciences, 2025                            | Tree-structured reasoning integrating KG and tabular data.                               | Outperforms table-only QA; limited by manual rule creation and low transparency in reasoning paths.                             |
| 7    | Reinforcement-Based KG Reasoning with Rule-and-Reinforce Triples       | Applied Sciences, 2025                                         | Logic rule mining with reinforcement learning-based triple selection.                    | Increases reasoning accuracy; computationally expensive and less interpretable.                                                 |
| 8    | CrimeKGQA: A Crime Investigation System Based on KG RAG                | Kuok et al., Technical Report, 2025                            | GPT-4o with Neo4j GraphCypherQChain and LangChain pipeline.                              | Enables crime-related querying through KG; limited by dataset quality and dependence on accurate Cypher generation.             |
| 9    | Deep Temporal Multi-Graph Convolutional Network for Crime Prediction   | Wang et al., ER Conference, 2020                               | Multi-graph convolutional networks with temporal features.                               | Accurately predicts crime hotspots; requires dense, high-quality spatio-temporal data for performance.                          |
| 10   | Enhancing In-Context Learning of LLMs for KG Reasoning                 | Zhang et al., Applied Sciences, 2025                           | Logic-guided context selection and reinforcement-based learning.                         | Improves reasoning via context filtering; generalization across unseen KGs remains a challenge.                                 |

### B. Constraints in Ontology and Knowledge Graph Creation Using LLMs

This research proposed a valuable method for generating ontologies and KGs using large language models, however, the framework is entirely dependent on prompt design and human intervention with AI-generated content. The resulting ontologies provided by the model often fail to articulate complex hierarchical relations and semantic constraints, limiting their dependability in professions such as law enforcement. Additionally, the evaluation is qualitative and not scalable, restricting the model's potential for automated or large volume crime KGs.

### C. Incompleteness in RAG-Based Reasoning Integration

The RAG-based investigation gives conceptual clarity to RAG, but is unclear on empirical nuance. Most of the apps analyzed are prototypes without clear criteria to define and evaluate retrieval consistency or accuracy of reasoning. The review identifies differences in knowledge synchronization, contextual drift, and explainability in several retrieval-augmented

pipelines. These highlights of these limitations show that existing RAG frameworks struggle to incorporate structured and unstructured data for both work and learning in high stakes, data intensive environments such as digital forensics or crime analysis.

### D. Scalability Issues in Knowledge Graph Question Answering Systems

This paper provides helpful information on KGQA architectures, but all of the articles are bound to the healthcare domain and use static datasets as the basis for the system in question. These papers all have concerns about scalability when transferring their systems to more dynamic environments with regular updates of new data. They also did not discuss the concern of ambiguity in multi-source or cross-domain question answering and, by extension, the same concerns would apply to investigative systems with varied evidence types and variable case data.

### *E. Limitations of Hybrid SPARQL and Keyword Search Approaches*

The hybrid approach to SPARQL-keyword was successful in increasing precision but still has limitations with more complex sentence structures and implicit relationships. The rule-based mapping approach is limited in dealing with linguistic diversity or unclear meaning due to context. The keyword search portion cannot organize results for significance or relevance to meaning. The approach lacks configurability and can not support queries in real time, especially for high-volume searches as is typical with investigative or legal data systems, where I prioritize adaptability, scalability, and context.

### *F. Dependence on Structured Data in Table-Based QA Systems*

This research improved hybrid QA using table and KG integration, but it requires access to neatly structured, clean tabular data for models to function competently. The tree-structured reasoning module still relies on manual decomposition rules, limiting utility in a domain. Furthermore, the method does not have a transparency-for-reasoning throughput; the user cannot follow how each component of an answer was produced, also making usage unsuitable in high-venue accountability contexts for work (i.e., criminal investigation).

### *G. Complexity and Opacity in Reinforcement-Based KG Reasoning*

The study by the Applied Sciences (2025) on the triple selection process that utilizes a reinforcement-learning method enhances accuracy of reasoning, at an expense of increased computation burden and complexity of training. Its reliance on logical rules that are pre-mined minimizes flexibility when applying it to new domain areas. Furthermore, the feedback based on reinforcement presents with a lack of interpretability, limiting knowing the precise reasons for giving priority to specific triples. The non-interpretability presents challenges for explainability and audit in these domains, which are important in forensic and judicial AI applications.

## IV. SCOPE OF RESEARCH

From the observed limitations, this section outlines promising research directions to achieve efficient, trustworthy, and domain-specific AI systems developed for crime analysis. Several limitations relate to enabling multi-hop reasoning, scalability of data integration, transparent retrieval pipelines, and interpretability of decision-making frameworks as advancements. These are ways to ensure AI-supported investigative systems in the future are more than experimental prototypes and can be operational tools utilized to substantiate evidence-based and legally defensible crime analysis.

### *A. Advancing Multi-Hop and Explainable Reasoning in KGQA*

In the future, research should focus on multi-hop reasoning models that can traverse complex relationships among entities and events in knowledge graphs. Providing explainable AI

modules in conjunction with graph traversal algorithms will allow a user to follow how each conclusion was reached. Improved entity-linking techniques and processes to incrementally update the graph will also contribute to the precision and flexibility of investigation systems while analyzing the changing data sets, thus generating trustworthy insight for difficult real-time analyses of criminal intelligence.

### *B. Improving Domain-Specific Ontology Generation with LLM Guidance*

Future efforts need to enhance ontology and KG generation from domain-tuned language models that have been trained on legal and investigative text corpora. Modern human-AI validation frameworks should be implemented to help in ensuring both semantic constancy and completeness with respect to the class hierarchy. Moreover, feedback-informed prompt optimisation and evaluation measures focused on completeness and correctness can greatly improve the automation as well as facilitate the AI models to remain interpretable at each step in the process. These advancements will help support and construct structured and high-quality crime knowledge graphs, which are critical to enabling more complex reasoning and contextualised decision support.

### *C. Developing Hybrid Retrieval-Augmented Pipelines for High-Stakes Domains*

To address and improve in RAG limitations, future systems should consider embedding symbolic reasoning with negligible retrieval and fact checking found in graphs. In addition, adaptive retrievers that can adaptively balance semantic (full-text) retrieval and structured retrieval (entities, etc.) will generate more factual relevancy. Retrieval logs, transparency, and parallel trust score calibration are important elements of trustworthiness reliability. These modern structures can ensure that an LLM content evaluation course of action gives correct information in an investigation, all while maintaining a large good content, court presentable representation.

### *D. Achieving Scalability and Real-Time Adaptation in KGQA Systems*

The evolution of KGQA systems will need them to support ingestion of incoming data in real time, cross-domain interoperability of knowledge graphs, and meaningful updates to dynamic knowledge graphs. Modular architectures takes advantage of graph databases and streaming APIs can increase the potential for scaling these knowledge bases. Future research should also involve reducing ambiguity in multiple-source data with contextual embeddings and entity disambiguation techniques. All together, these applications would permit knowledge-based investigative assistants to process developing evidences in real time and parallelly maintain accuracy and temporal relevance across various crime data sets.

## V. CONCLUSION

This review looked at how intelligent crime investigation systems are evolving, especially those that combine large

language models (LLMs) with knowledge graph reasoning. This studies reviewed show that hybrid AI systems—which mix symbolic, neural, and retrieval-based methods—can make investigations more accurate, clear, and meaningful.

However, there's still a lot of work to be done. These systems need to become more scalable, transparent, and adaptable to handle real-time data from multiple sources. Right now, most of them are still in the experimental stage, using static datasets, taking a long time to process, and requiring high computing power.

In the future, researchers aim to create LLMs that are better suited for crime investigation, combining their strong language understanding with structured investigative data. The goal is to make these systems capable of updating information and reasoning transparently.

Further research will focus on human involvement in verification, smarter information retrieval, and multi-step reasoning using graphs. These improvements will help make AI systems more reliable and accountable.

Ultimately, the next generation of AI tools for law enforcement will focus on connecting and interpreting evidence more effectively. Success will depend on balancing innovation with clarity, trust, and usability, so that these systems can be easily used by police and investigators in real-world cases.